



EEE 5776: Applied Machine Learning I

Course Overview

Professor:

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EEL 5776

APPLIED MACHINE LEARNING

This course provides a comprehensive introduction to machine learning and deep learning, focusing on practical applications and hands-on implementation. Students will explore the machine learning pipeline, including data preprocessing, model development, evaluation, and deployment. The course covers both supervised and unsupervised learning techniques, such as regression, classification, clustering, and dimensionality reduction, alongside advanced deep learning methods, including Artificial Neural Networks (ANNs), Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs). By the end of the course, students will have developed the skills to build, evaluate, and present machine learning and deep learning solutions using state-of-the-art tools and frameworks



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Upon completion of this course, students will be able to:

- Utilize terminology for Artificial Intelligence (AI), Machine Learning (ML) and Deep Learning (DL) tools.
- Design and conduct meaningful experiments to evaluate the performance of ML models.
- Determine which ML model to use for an application and/or task.
- Identify and explain strengths and limitations of ML models.
- Select appropriate metrics of success.
- Implement in code several ML models utilizing state-of-the-art off-the-shelf libraries.

Course objectives

Andrea Ramirez-Salgado, Ph.D.

Instructional Assistant Professor @ Department of Engineering Education

Office Hours:

Tuesdays and Thursdays 10-11am

MALA4112

Contact:

(1) Slack, (2) Email: aramirezsalgado@ufl.edu

Topics:

AI+IoT (AloT), edge AI, intelligent embedded systems

Research:

Teaching and learning approaches to enhance the representation of neurodiverse and underrepresented populations in computer engineering

Previous work:

Undergrad and grad courses in CS, ECE, and curriculum and instruction +
Research in inclusive and equitable learning environments + Civil engineering company

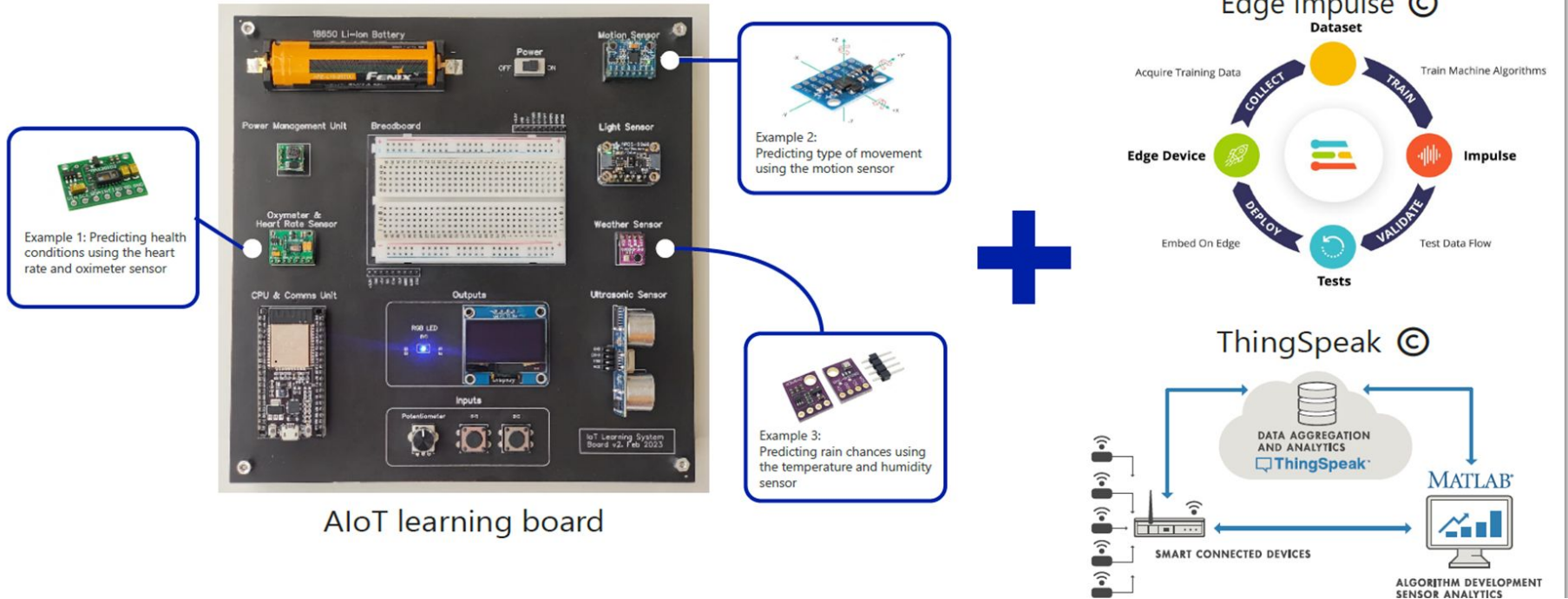


About me

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Curricular Interest – Edge AI and AIoT - Core Member of the [Nelms Institute for the Connected World](#)

About me

Padlet



- Your name
- Where you are you from
- Education background
- Anything specific you want us to know about you/ fun fact

Things I want to know about you



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● Canvas.

- Canvas is our principal hub
- Here I will post the resources, assignments and deadlines
- All deadlines will be on Sundays at 11:59pm (one exception detailed later)
- Inbox

Spring 2025

Home

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EEE5776 - Applied Machine Learning

Assign To Edit

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APPLIED MACHINE LEARNING

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Welcome Schedule Modules

General Information and Resources

- ✓ Module 1. Introduction to Machine Learning
- ✓ Module 2. Supervised Learning and Experimental Design
- ✓ Module 3. Unsupervised Learning
- ✓ Module 4. Introduction to Deep Learning and Neural Networks

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Course structure

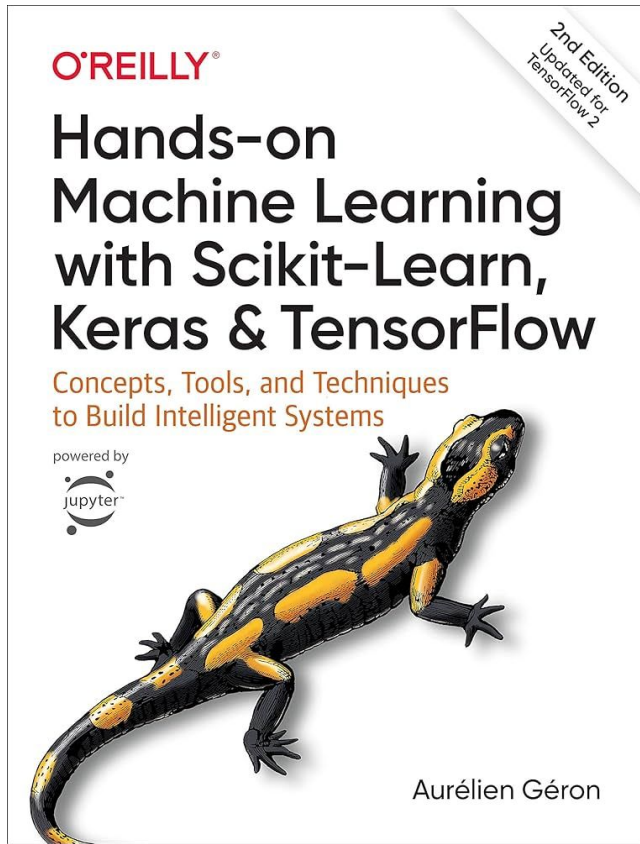


- **GitHub.**
 - Canvas is our secondary hub
 - Here I will post the jupyter notebooks, homeworks and project repositories



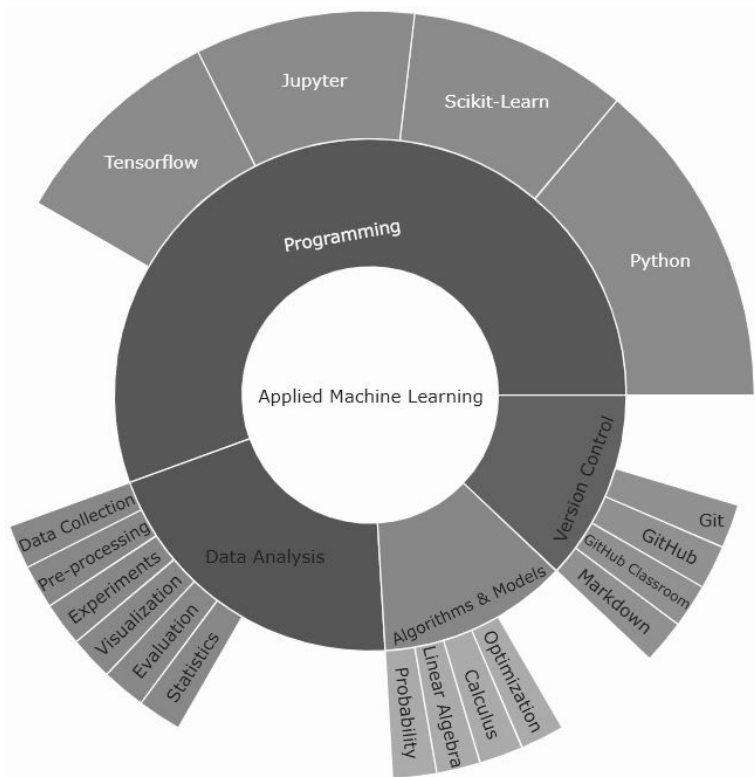
- **Slack.** [uf-eee5776-AppliedML-Spring25](#)
 - This resource is optional and intended to supplement office hours and student interaction
 - No official communication/submission happens over Slack

Course structure



- **Main Textbook**
 - **Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow.** Aurélien Géron. O'Reilly Media, 2019
 - Available as electronic copy with Course Reserves (in Canvas)
 - All materials and slides will be available in Canvas

Materials



- Statistics (e.g., CLT, hypothesis testing, estimators)
- Basic probability theory (e.g., Gaussian and discrete RVs, probabilistic models)
- Programming (Python preferred but not required)

This course will be taught under the assumption that a student does not have experience programming in Python.

Python will be presented in the context of machine learning.

Tools to teach ML

The logo for Conda, featuring the word "CONDA" in green capital letters with a green snake-like icon to the left.The matplotlib logo, featuring the word "matplotlib" in blue lowercase letters with a small orange and blue icon to the right.

- **Anaconda Distribution:**

- <https://www.anaconda.com/download>

- Includes Python 3.12
 - It includes all libraries, modules and tools we will use: Jupyter Notebooks, NumPy, Matplotlib, SciPy, Pandas, Scikit-Learn, TensorFlow

- **Install it (or update it) before next lecture**

Software

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- You have 2 options to manage your packages and virtual environment/s:
 - **using pip.** System that manages Python packages.
 - **using conda.** System that manages packages that may be written in any programming language.
- Since we will use Python packages, you can use either one of these systems to manage your virtual environment. Which one to use typically comes with your specific needs. Finding help for creating and managing your conda environments:
 - [using conda to manage conda environments.](#)
 - [using pip to manage conda environments.](#)

Package Managers: conda & pip



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Week	Date	Module	Topic/s
1	1/14	Introduction to Machine Learning	Course overview, expectations, and goals. Types of learning: Supervised, Unsupervised, and Reinforcement Learning. The machine learning pipeline: Raw data → Preprocessing → Model → Evaluation.
	1/16		Introduction to Jupyter Notebooks, NumPy, Matplotlib, Pandas, and Scikit-Learn.
2	1/21	Supervised Learning and Experimental Design	Introduction to tasks: Regression (Linear Regression), Classification (Logistic Regression). Basic pipeline setup with Scikit-Learn: Input data → Model → Prediction.
	1/23		Data cleaning: Handle missing values and outliers. Feature scaling and encoding categorical variables (practical). Hands-on integration: Use cleaned data to train and evaluate models.
3	1/28		HiPerGator info session.
	1/30		Hyperparameter tuning and sampling strategies (cross-validation, nested CV, stratified CV, Bootstrap). Hands-on integration: Train models with cross-validation.
4	2/4		Introduction to supervised learning for classification tasks. Logistic Regression
	2/6		Model selection and performance metrics (hypothesis testing, confidence intervals, ROC, F1, etc.).
5	2/11		Regularization.
6	2/13		Decision Trees. Random Forests.
	2/18		Bagging. Boosting.
7	2/20		Gradient Boosting Machines (GBM).
	2/25		Hard-Margin Support Vector Machines. Slack variables.
	2/27		Soft-Margin Support Vector Machines.

8	3/4	Midterm Exam	
	3/6	Unsupervised Learning	K-Means Clustering. Introduction to DBSCAN. Clustering validity measures.
9	3/11		The Curse of Dimensionality. Principal Component Analysis (PCA).
	3/13		K-Nearest Neighbors.
10	3/18 3/20	Spring Break (No class)	
11	3/25	Introduction to Deep Learning	Introduction to TensorFlow. Artificial Neural Networks. Backpropagation.
	3/27		Best practices for training artificial neural networks I
12	4/1		Best practices for training artificial neural networks II
	4/3		Convolutional Neural Networks (CNNs) I
13	4/8		Convolutional Neural Networks (CNNs) II
	4/10		Recurrent/Sequential Models I
14	4/15		Recurrent/Sequential Models II
	4/17		Explainable Deep Learning
15	4/22		Fairness in Deep Learning. Final Exam Review
NA	4/29	Final Exam @10:00a-12:00p	

Course schedule



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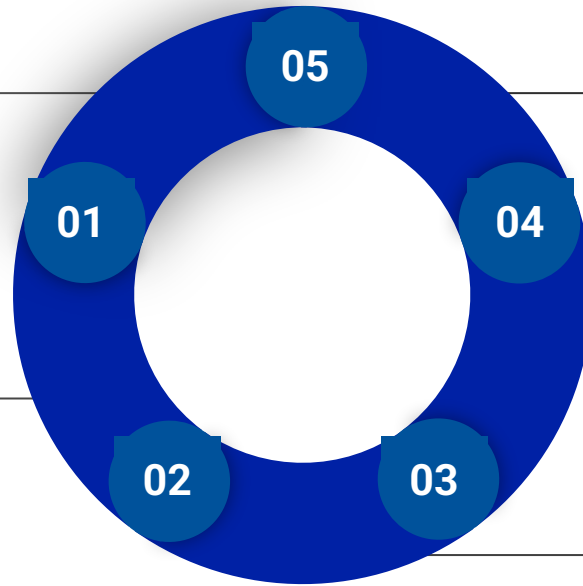
Assignment	Total Points	Percentage
Exam 1	100	20 %
Exam 2	100	20 %
Project 1	100	15 %
Project 2	100	15 %
Project 3	100	15 %
Poster Presentation	100	5 %
Homework	100	10 %
		100%

- **Homework:** will consist of practical and theoretical understanding of the topics covered in class. A typical homework will have two components: Part I – consists of a quiz that will assess theoretical understanding; Part II – consists of practical problem/s to be implemented in Python.
- **Exams:** (1) The exams will be drawn evenly from all lectures, assignments, and readings that occurred up to that point in the course. (2) Exams will have 2 parts: Part I – theoretical questions; part II – simulation questions to be solved using Python and Jupyter Notebooks.
- **Projects:** All projects are individual assignments. For each project, students are expected to write a report, submit their code and create a written demo (README file) on how to use their code
- **Poster presentations** will showcase the individual projects, grouped by their respective project themes.

Grading

**Interdisciplinary
skill
development.**

**Project-based
learning**



**Cutting-edge AI
expertise**

**Real-world
application**

**Industry
relevance**

Why is this important



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- I will prepare course lectures with the expectation that **students will attend class synchronously** and bring a computer to follow along with any practical implementations.
- Makeups for exams, homework assignments, and weekly assignments are **NOT normally allowed**. If you cannot attend an exam, you must contact the instructor well in advance (at least 7 days before an announced exam date).
- Unless otherwise noted, there are **no group or collaborative assignments** in this course.
- If you're unsure about the acceptable use of AI tools for your work, **ask for clarification!**
- Students will complete this course with honor and integrity, or not at all. Submissions which are believed to be not entirely a student's own work will be **reported to administration** for disciplinary action. In ALL cases, I will recommend the following sanctions be imposed on that student or students:

A failing grade (an 'E') for the course

That you not be allowed to drop the course for any reason

- Late submissions will not be accepted.



**Read the
syllabus!**

Course Policies



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✓ As your professor, I will:

- Come to class prepared
- Strive to use class time wisely by structuring relevant and engaging activities
- Avoid busywork by relating assignments to our classroom sessions & practical applications of concepts
- Take into account student feedback
- Recognize that learning is a partnership between instructor & student

✓ As the student, you should:

- Recognize that learning is a partnership between instructor & student
- Acknowledge the difference between being a “student” and being a learner
- Engage in activities & be “present” in class
- Be respectful to everyone in our community
- Communicate with me immediately should any problems arise
- Appreciate the important role you play in how valuable any course is to you

Tips to be successful

Graduate School

<https://graduateschool.ufl.edu/>

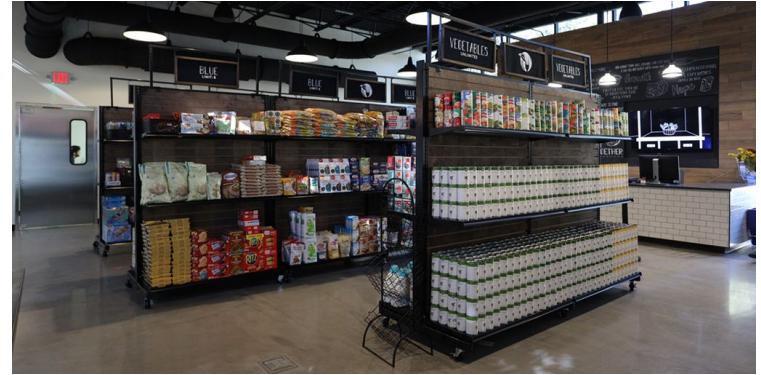
For example:

Alan and Cathy Hitchcock Field & Fork Pantry

<https://pantry.fieldandfork.ufl.edu/>

Lake Wauburg

<https://recsports.ufl.edu/locations/lake-wauburg/>



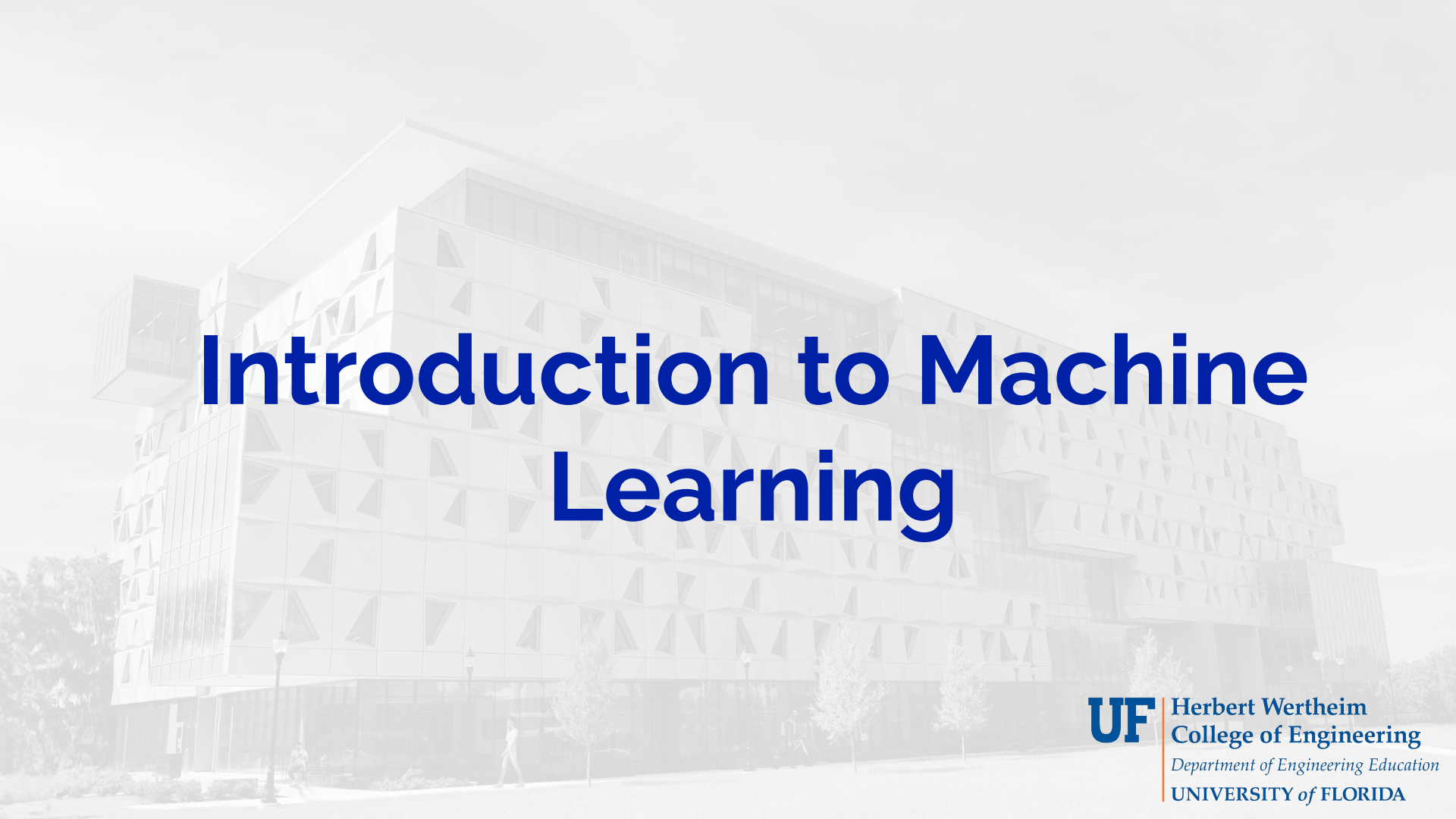
Off the shelf - UF Resources

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Any Questions?

Let's get started!



Introduction to Machine Learning



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What is AI?

Artificial Intelligence, or AI, refers to systems that **simulate human intelligence** to perform tasks and improve over time through experience and data

What should be the key capabilities of AI if the purpose is simulating the human intelligence?

Perception

Reasoning

Decision-making

Natural Language
Understanding

AI Overview

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Programming vs. AI



**Traditional
Programming**



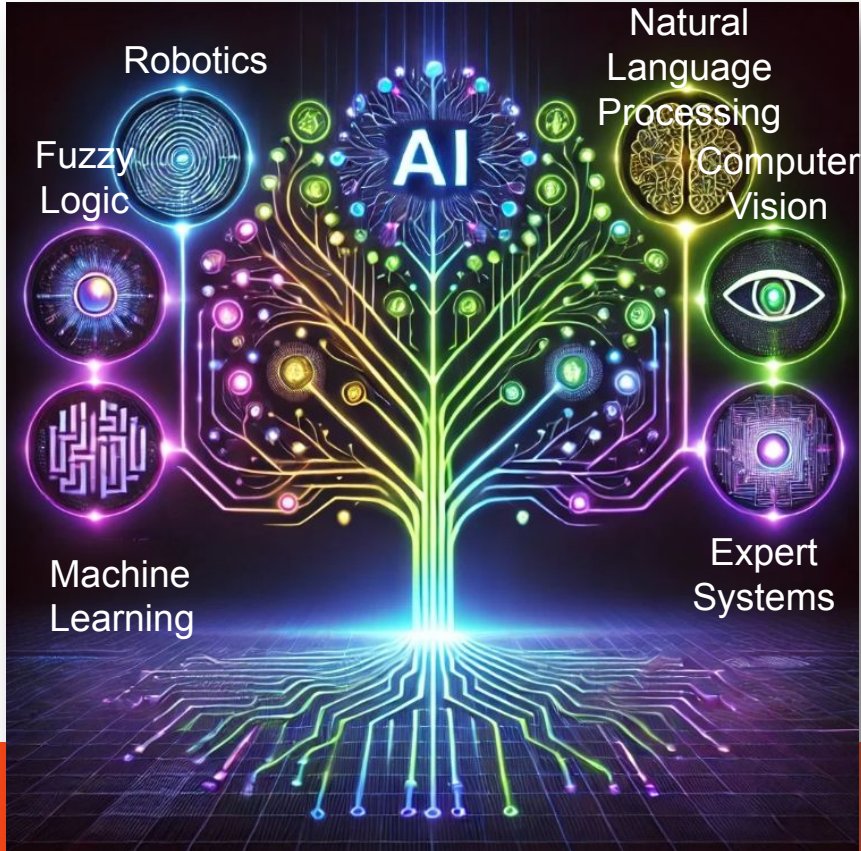
**Artificial
Intelligence**

AI Overview

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AI Branches



Where are the boundaries between all these branches?

AI-generated image using DALL-E.
Prompt: A futuristic tree with seven nodes. The main one is AI, three at the left and three at the right of the main node. The nodes represent AI branches



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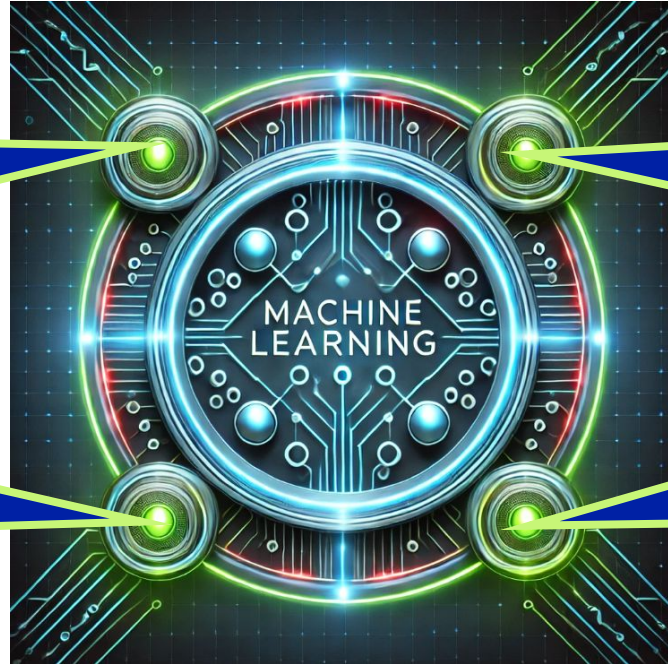
Machine Learning

Unsupervised
Learning
models

Supervised
Learning
models

Reinforcement
Learning

Deep
Learning



AI-generated image using DALL-E. Prompt: A futuristic circle with four external nodes in the edge and the label "machine learning" in the inner circle

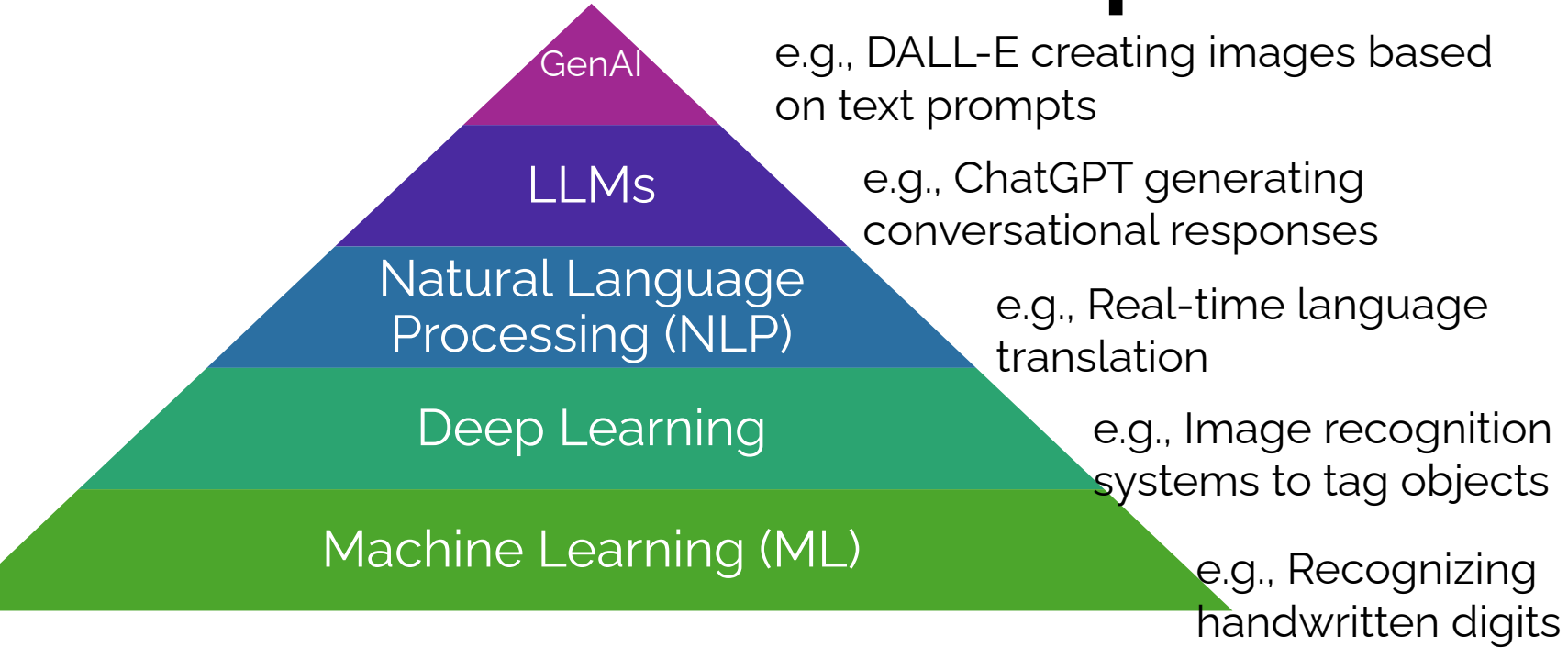
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AI Branches amplified



AI Overview

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Supervised Learning - The Big Picture

- **Analogy time!**

- Supervised learning is like studying vocabulary with flashcards. Each card shows a word (**input**) on one side and its meaning (**label**) on the other. Over time, the learner associates words with their meanings and can understand them in new contexts

Supervised Learning Tasks



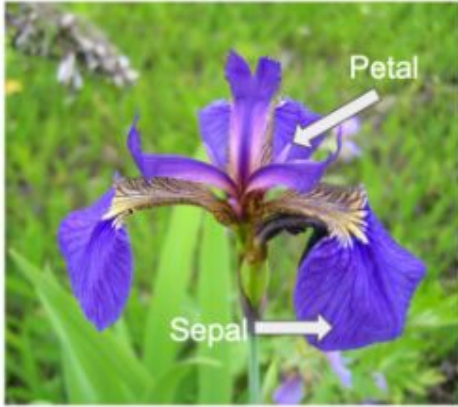
Supervised Learning

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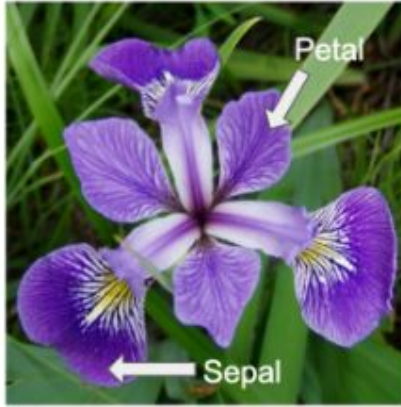
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Supervised Learning: Classification Example

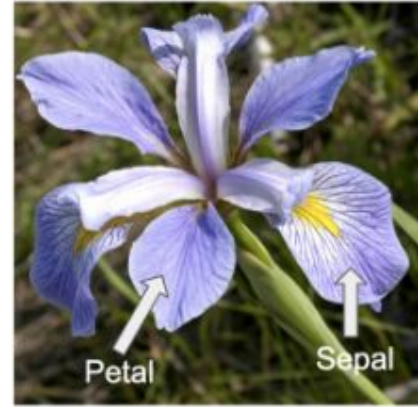
- **Example:** Iris Species Flower Classifier Using the Sepal and Petal Width and Length in cm. Public dataset containing 150 samples



Setosa



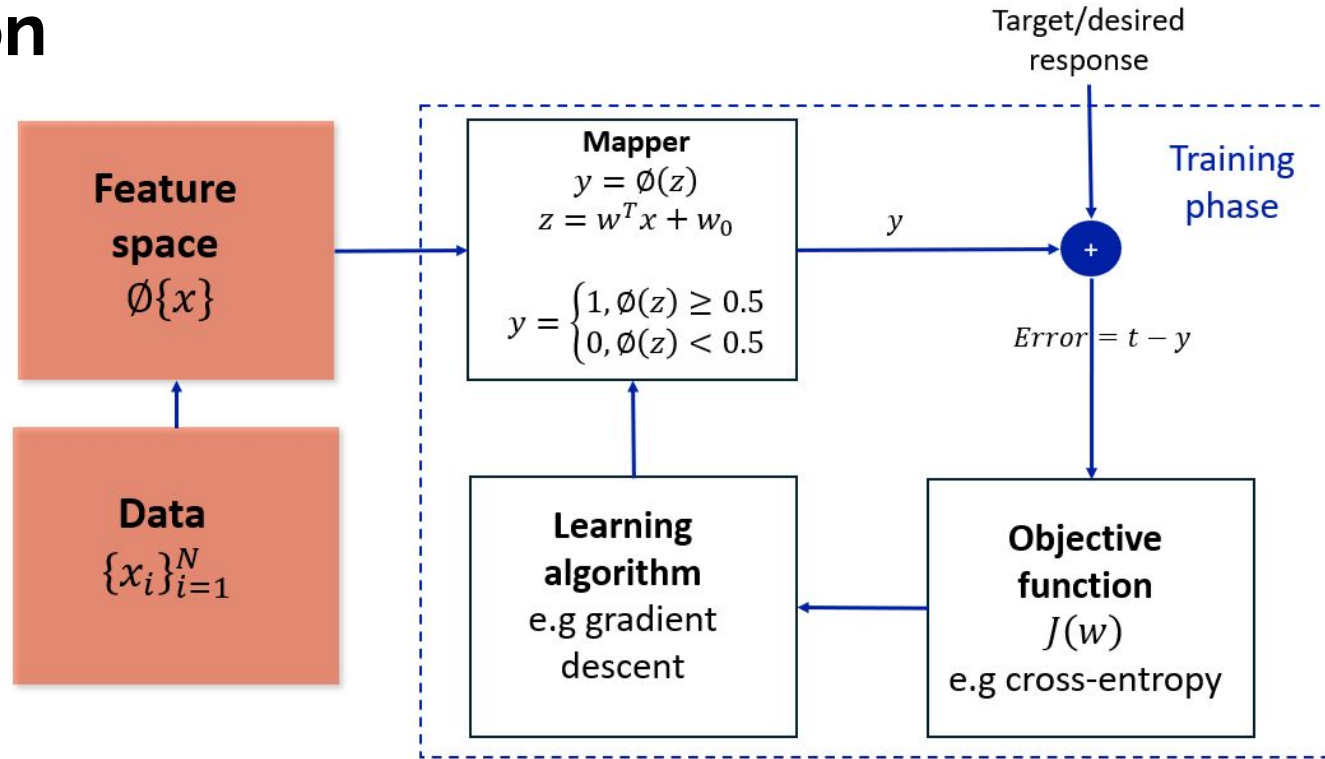
Versicolor



Virginica

Supervised Learning

Supervised Learning System: Logistic Regression



Supervised Learning



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Supervised Learning - Regression

- **Example:** Predicting House Prices in California Using a Public Dataset

- **Features:**

`MedInc', 'HouseAge', 'AveRooms', 'AveBedrms', 'Population',
'AveOccup', 'Latitude', 'Longitude'`

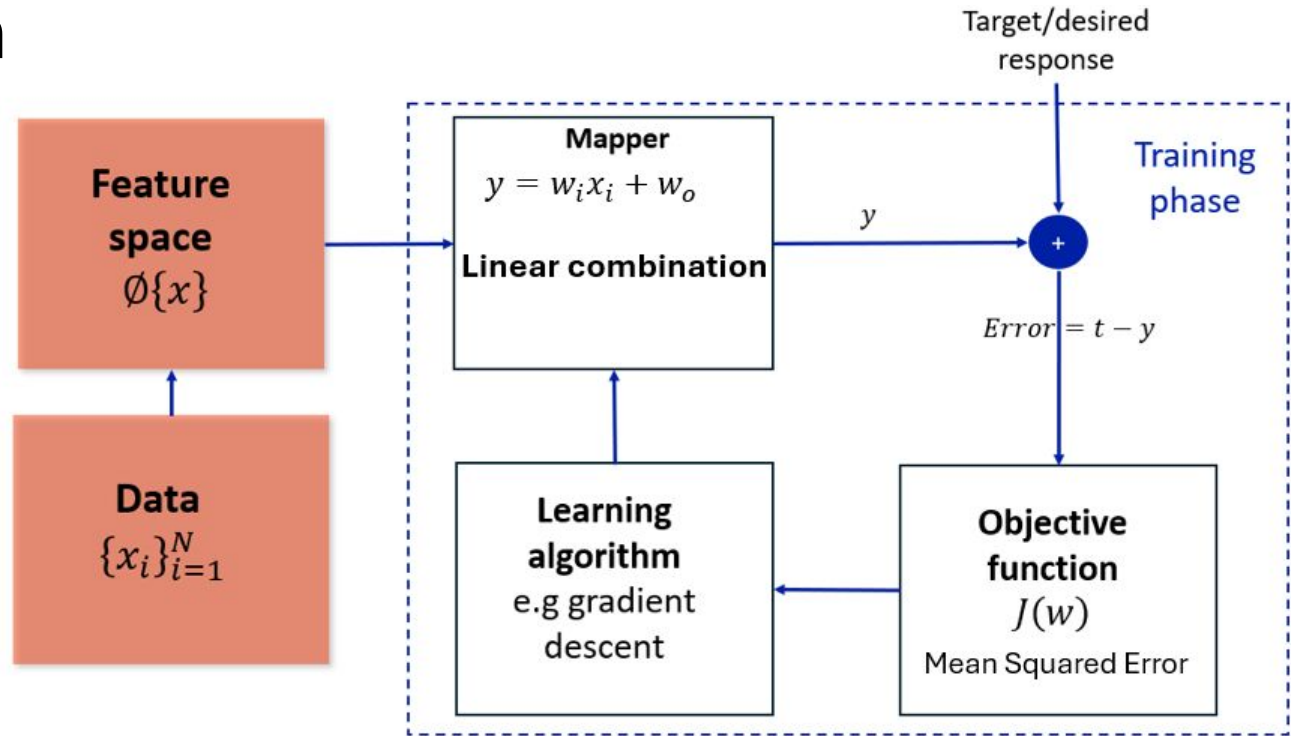


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Supervised Learning System: Linear Regression



Supervised Learning



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Supervised Learning: Regression Example

**Let's work with this
inputs**

Median income of \$60k
Average house age of 20 years
Average 5 rooms per house
Average 1 bedroom per house
1000 people in the block
Average 3 occupants per
household
Latitude: 34
Longitude: -118

What is your price prediction?

Supervised Learning



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Supervised Learning: Regression Prediction

Predicted Median House Value for Input Data: \$292178.36

Model Evaluation:

Mean Squared Error: 0.5558915986952444

R² Score: 0.5757877060324508

Feature Importances (from model coefficients):

Average Bedrooms	0.783145
Median Income (in \$10k)	0.448675
Longitude	0.433708
Latitude	0.419792
Average Rooms	0.123323
House Age (in years)	0.009724
Average Occupancy	0.003526
Population in Block	0.000002
dtype: float64	

Supervised Learning

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Unsupervised Learning – The Big Picture

- **Analogy time!**

Unsupervised learning is like trying to organize the pieces of a jigsaw puzzle without seeing the final image. You group pieces based on colors, patterns, or edges, gradually discovering relationships and forming clusters, even though you don't know the complete picture

Unsupervised Learning Approaches



Clustering

Dimensionality
Reduction

Unsupervised Learning

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Unsupervised Learning – Clustering

- **Example:** Suppose you collect pictures of the following objects.
How many groups would you partition this data into?



Unsupervised Learning

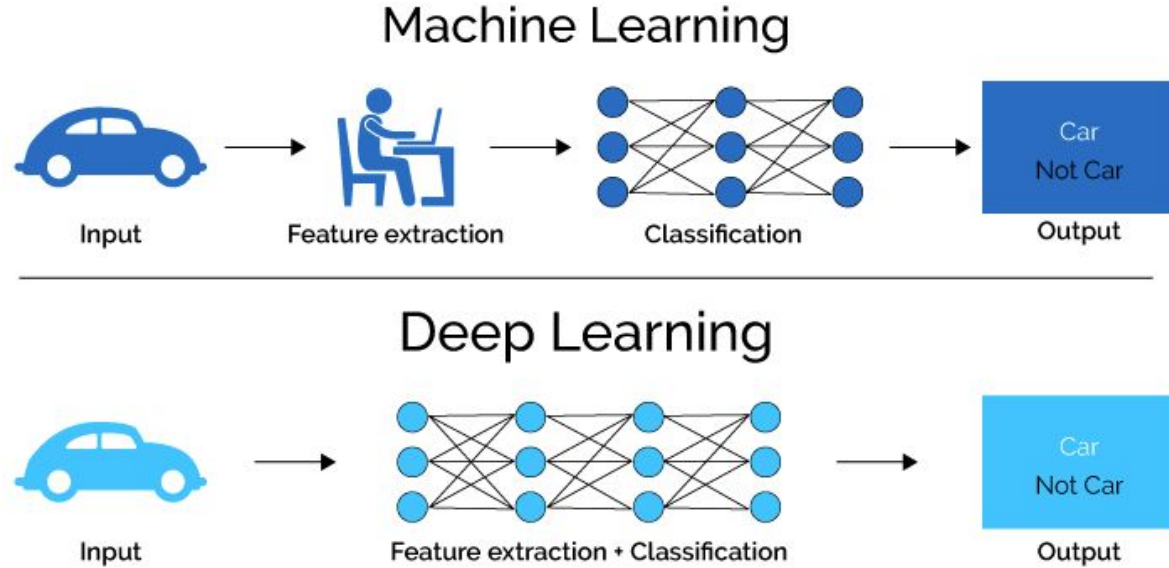
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Deep Learning : The Big Picture

- Analogy time!

- Deep learning is like a group of archaeologists digging through layers of earth to uncover an ancient city. Each layer reveals more clues and details, eventually piecing together the full structure.

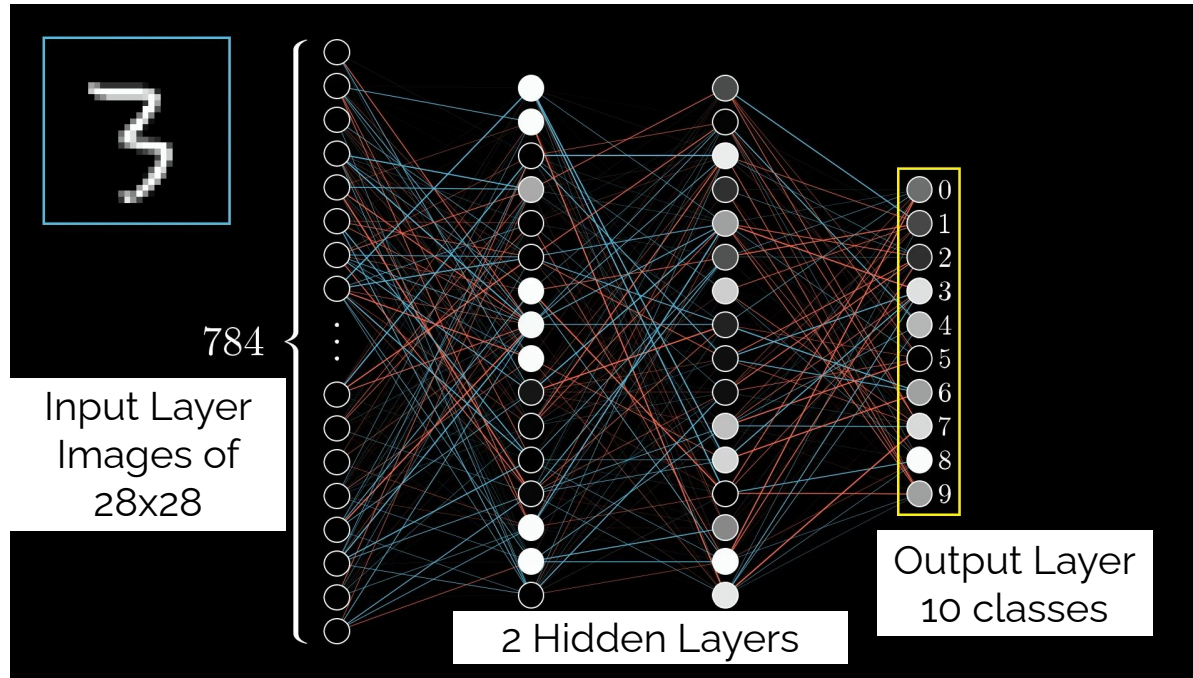


Deep Learning

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Deep Learning: The Big Picture



Source:
3Blue1Brown

Deep Learning

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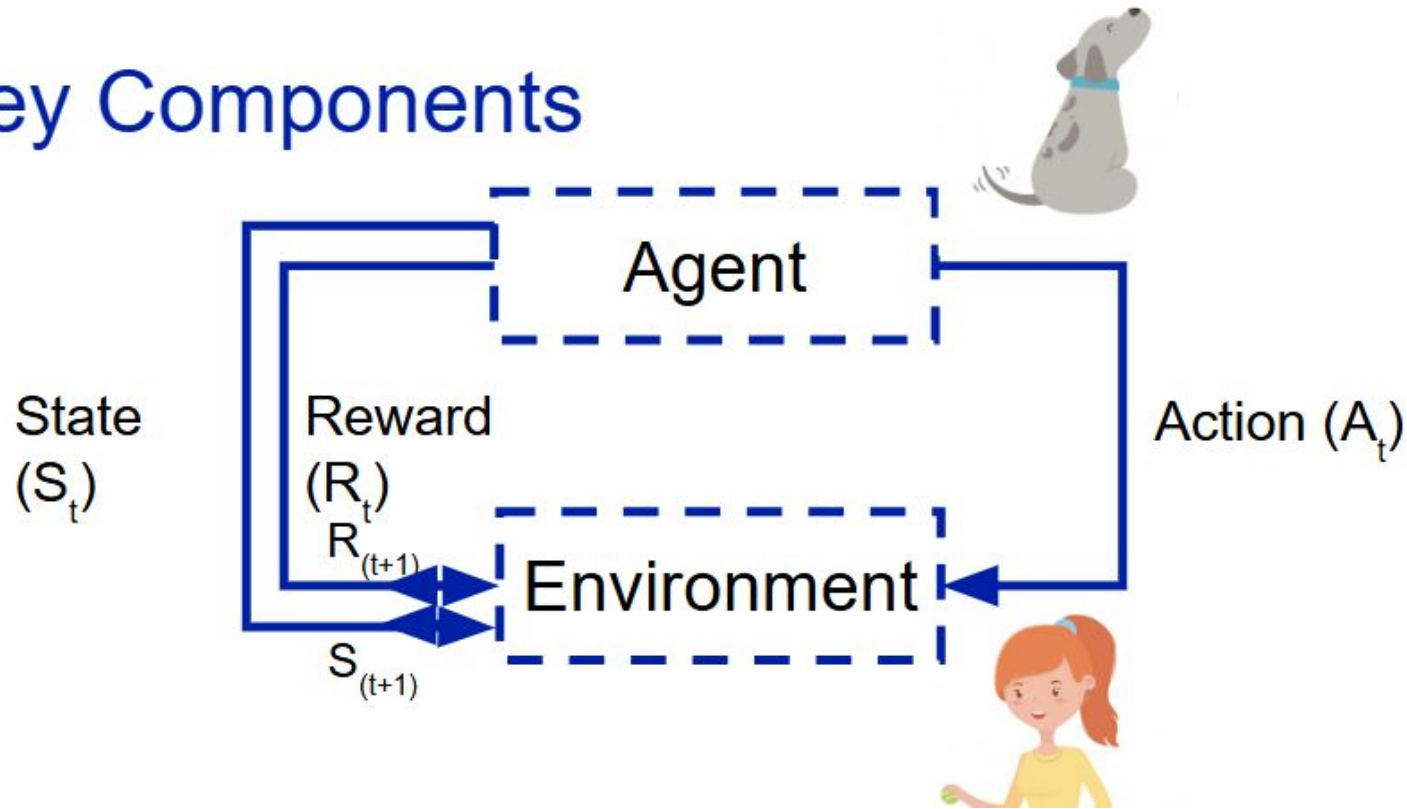
Reinforcement Learning – The Big Picture

- Analogy time!

- Reinforcement learning is like training a puppy: rewarding it with treats when it fetches correctly and withholding treats when it doesn't. Over time, the puppy learns the desired behavior.



Key Components



Reinforcement Learning



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Large Language Models (LLMs), Generative AI, and NLP

- Analogy time!

NLP is like a translator who deciphers human language and extracts its meaning

LLMs are like a master librarian who has read millions of books. They can retrieve and recombine information to answer your questions or create coherent texts.

Generative AI is like an artist who takes your rough description and creates something unique—whether it's a story, an image, or a piece of music

Generative Models



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